

[Learn](#) [Practice](#)

Grade 11 Chemical Bonding and Hybridisation — Issue #36 Candidate B:

Source questions

Q1 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Delocalised pi systems and stationary resonance hybrids

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Delocalised pi systems in planar oxyanions

Question type: Constructed response

Source: Owner-supplied question

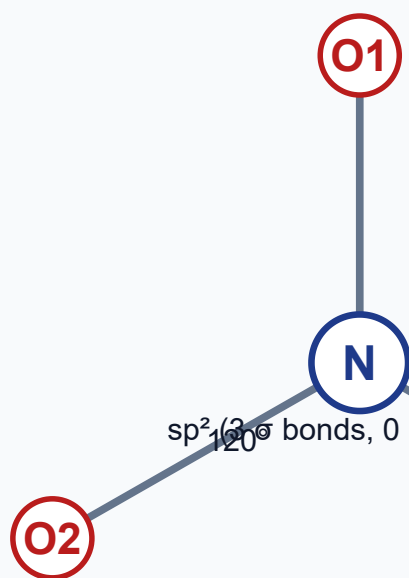
Provenance: Owner-supplied question

ATTEMPT FIRST

Nitrate, NO_3^- , is planar and has three equivalent N–O bonds. Its three conventional Lewis contributors place one N=O bond in different positions. Choose between a permanently localised single/double-bond picture and a delocalised pi description. Explain which observations discriminate between them, and state separately the introductory sigma-framework hybridisation model at nitrogen.

Representation

Stage 1: Trigonal Planar Sigma Framework



Planar sigma frame

**Common wrong route •
open if you want a
warning**

—

Common wrong route

Equivalent bonds are a
permanently selected
Lewis double bond.

Need a hint? • guided support

—

HINT LADDER • REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Separate supplied bond
equivalence from a
permanently chosen
Lewis double bond.

H1 Representation

Guided

Equivalent contributors
are bookkeeping
alternatives for one
delocalised description.

H2 First move

Guided

Need the concept again?

Delocalised pi systems and stationary
resonance hybrids

Compare what a fixed unequal-bond model predicts with the supplied equivalent positions.

Answer and working

Hide

Reasoning route

1 Understand

Compare a fixed localized nitrate picture with the supplied equivalence.

Uses

Nitrate, NO_3^- , is planar and has three equivalent N–O bonds. Its three conventional Lewis contributors place one N=O bond in different positions. Choose between a permanently localised single/double-bond picture and a delocalised pi description. Explain which observations discriminate between them, and state separately the introductory sigma-framework hybridisation model at nitrogen.

Why valid: A permanently unique N=O position distinguishes one bond; the observation instead supplies three equivalent N–O bonds.

Result: The fixed single/double-bond assignment is inadequate.

2 Connect

Key move

Choose one delocalised pi account for all contributors.

Uses The fixed single/double-bond assignment is inadequate.

Why valid: The contributors are alternative bookkeeping descriptions of one electronic state, not switching structures.

Result: One delocalised pi system spans the N–O framework.

3 Calculate

Specify nitrogen sigma accounting independently.

Uses One delocalised pi system spans the N–O framework.

Why valid: Three planar local sigma directions support the introductory sp^2 framework; equivalence alone is not its criterion.

Result: Three N sigma directions; residual perpendicular p participates in the pi space.

4 Interpret

Test the equivalence-implies-sp² inference on NH₃.

Uses Three N sigma directions; residual perpendicular p participates in the pi space.

Why valid: NH₃ can have three equivalent N-H bonds while having four local domains.

Result: Bond equivalence does not uniquely select an sp² description.

Planar equivalent nitrate bonds reject one permanently unique N=O position. N sigma framework is locally sp²; equivalence alone is not the hybridisation criterion.

Independent check

Changed check: supplied equivalent CO₂ bonds coexist with sp carbon; equivalence cannot alone imply sp².

Test the rule on a changed case

Can equivalent bonds at CO₂ carbon and nitrate nitrogen imply the same local hybrid label? Recount directions.

Your prediction and reason

Q2 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Delocalised pi systems and stationary resonance hybrids

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Delocalised pi systems in planar oxyanions

Question type: Constructed response

Source: Owner-supplied question

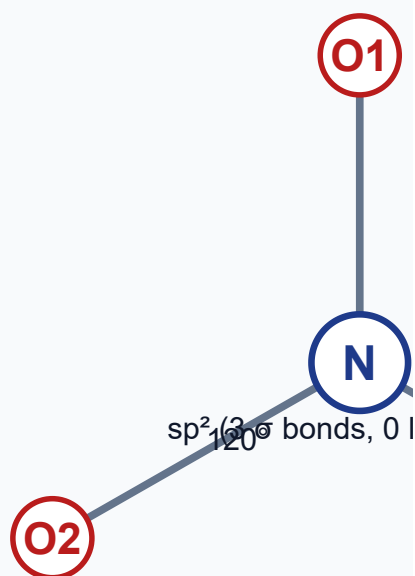
Provenance: Owner-supplied question

ATTEMPT FIRST

Use the supplied nitrate description: three planar sigma bond directions around nitrogen and a pi system delocalised across the N–O framework. Translate the three Lewis contributors into one qualitative orbital diagram. Distinguish the sigma framework from the delocalised pi system, and explain what is preserved and what must not be interpreted as a molecule switching between drawings.

Representation

Stage 1: Trigonal Planar Sigma Framework



Planar sigma frame

Common wrong route •
open if you want a
warning

Common wrong route

Resonance contributors are molecular structures switching in time; only the drawn double bond supplies pi electrons.

Need a hint? • guided support

—

**HINT LADDER • REVEAL
ONLY WHAT YOU NEED**

Guided support

H0 Key concept

Guided

Translate formal contributors into one sigma framework and a complete pi-space ledger.

H1 Representation

Guided

A formal Lewis double bond does not identify the whole delocalised occupancy.

H2 First move

Guided

List the central and terminal perpendicular p contributions and preserve one set of nuclear positions.

H3

Equation / formal model

Guided

Test your proposed route on this changed case

Need the concept again?

Delocalised pi systems and stationary resonance hybrids

before finalising: For nitrate and carbonate, count pi-space electrons including terminal perpendicular lone pairs. Do formal contributors describe alternating nuclear structures?

H4

Assembly checkpoint

Guided

Reserve the last judgement for yourself. Try this boundary/transfer task: Translate carbonate contributors into a continuous p-space description.

Answer and working

Hide

Reasoning route

1

Understand

Preserve the nitrate nuclei and sigma connectivity.

Uses

Use the supplied nitrate description: three planar sigma bond directions around nitrogen and a pi system delocalised across the N-O framework. Translate the three Lewis contributors into one qualitative orbital diagram. Distinguish the sigma framework from the delocalised pi system, and explain what is preserved and what must not be interpreted as a molecule switching between drawings.

Why valid: Every supplied contributor has the same N centre and three N-O connections.

Result: Three coplanar N-O sigma directions.

2**Connect**

Key move

Build a perpendicular p basis across all four centres.

Uses Three coplanar N-O sigma directions.

Why valid: A delocalised qualitative diagram combines the central N and three O p functions.

Result: Four perpendicular p functions spanning one pi space.

3**Calculate**

Count pi-space electrons from one contributor.

Uses Four perpendicular p functions spanning one pi space.

Why valid: One N=O pi pair and p-oriented pairs on the two singly bonded O sites supply three pairs.

Result: Six pi-space electrons; the Lewis double bond alone is not the complete pi inventory.

4**Interpret**

Check contributor invariants rather than a switching animation.

Uses

Six pi-space electrons; the Lewis double bond alone is not the complete pi inventory.

Why valid: Changing the drawn N=O position changes bookkeeping, not nuclei, total charge or electron number.

Result: One planar equivalent-bond state; no molecule oscillating between three structures.

Nitrate has three in-plane N-O sigma directions and four perpendicular p functions spanning a six-electron pi space. Contributors do not switch in time.

Independent check

Changed check: 24 total valence electrons = 18 in sigma/in-plane lone-pair bookkeeping plus six in the perpendicular pi space.

Test the rule on a changed case

For nitrate and carbonate, count pi-space electrons including terminal perpendicular lone pairs. Do formal contributors describe alternating nuclear structures?

Your prediction and reason

Q3 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Cumulated pi bonding and allene stereochemical orthogonality

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Cumulated polyene stereochemistry and orbital orthogonality

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For propadiene (allene), $\text{H}_2\text{C}=\text{C}=\text{CH}_2$, apply local electron-domain counting at each of the three carbon atoms. Assign the introductory hybridisation of each carbon and show how the two mutually perpendicular unhybridised p orbitals on the central carbon constrain the

Representation

Allene connectivity: $\text{H}_2\text{C}=\text{C}$
Construct directions; do not

Connectivity

relative planes of the terminal
CH₂ groups.

**Common wrong route •
open if you want a
warning**

—

Common wrong route

All double-bonded
fragments must share
one plane.

Need a hint? • guided support

—

**HINT LADDER • REVEAL
ONLY WHAT YOU NEED**

Guided support

H0 Key concept

Guided

Assign local bases before
deducing the relation
between terminal planes.

H1 Representation

Guided

A central sp carbon
retains two perpendicular
p directions.

H2 First move

Guided

Need the concept again?

Cumulated pi bonding and allene
stereochemical orthogonality

Match each central p direction with an aligned terminal p direction; infer terminal sigma planes yourself.

H3

Equation / formal model

Guided

Test your proposed route on this changed case before finalising:

Compare ethene with allene: how many residual p directions are needed at the centre, and which planes must align with each?

H4

Assembly checkpoint

Guided

Reserve the last judgement for yourself.

Try this

boundary/transfer task:

Build the allene end-on projection from the two p directions.

Answer and working

Hide

Reasoning route

1 Understand

Count local sigma-neighbour directions in allene.

Uses

For propadiene (allene), $\text{H}_2\text{C}=\text{C}=\text{CH}_2$, apply local electron-domain counting at each of the three carbon atoms. Assign the introductory hybridisation of each carbon and show how the two mutually perpendicular unhybridised p orbitals on the central carbon constrain the relative planes of the terminal CH_2 groups.

Why valid: Each terminal carbon has two H plus the central C; the central carbon has two terminal-C neighbours.

Result: Terminals: three directions; centre: two.

2 Connect

Key move

Allocate the local valence bases.

Uses Terminals: three directions; centre: two.

Why valid: Terminal sp^2 leaves one p each; central sp leaves two mutually perpendicular p functions.

Result: sp^2 - sp - sp^2 along the carbon chain.

3 Calculate

Match the two separate central p directions to terminal p functions.

Uses sp^2 - sp - sp^2 along the carbon chain.

Why valid: Each localized pi component needs suitable lateral alignment with its terminal p.

Result: Two orthogonal pi planes.

4 Interpret

Deduce and check terminal CH_2 planes.

Uses Two orthogonal pi planes.

Why valid: Each terminal sigma plane is perpendicular to its participating p; looking down the chain makes the relation explicit.

Result: Terminal CH_2 planes are perpendicular in this introductory model; rigid rotation preserves that relation.

Allene uses terminal sp^2 , central sp accounting. The two central perpendicular p directions support perpendicular terminal CH_2 planes in the introductory

model.

Independent check

Changed check: looking down $C=C=C$, terminal CH_2 planes project at right angles; rotating the whole model leaves that relation unchanged.

Test the rule on a changed case

Compare ethene with allene: how many residual p directions are needed at the centre, and which planes must align with each?

Your prediction and reason

Q4 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Cumulated pi bonding and allene stereochemical orthogonality

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Cumulated polyene stereochemistry and orbital orthogonality

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Ethene is planar; propadiene has terminal CH_2 groups in perpendicular planes. Build a connected account from each carbon's local sigma framework, the required pi overlaps and the p-orbital

Representation

Allene connectivity: $H_2C=C$
Construct directions; do not

Connectivity

directions. Explain why copying the planar ethene sketch to allene fails.

Common wrong route • open if you want a warning

Common wrong route

Pauli exclusion forbids one atomic orbital from participating in any multicentre description.

Need a hint? • guided support

HINT LADDER • REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Explain the terminal-plane constraint using overlap geometry, not an invalid occupation claim.

H1 Representation

Guided

Atomic basis functions participate in molecular combinations; Pauli

Need the concept again?

Cumulated pi bonding and allene stereochemical orthogonality

exclusion is an electron-state rule.

H2 First move

Guided

Name the two inter-centre pi overlaps and compare their directions.

H3

Equation / formal model

Guided

Test your proposed route on this changed case before finalising: Does forming a pi contribution make an atomic p function unavailable to every multicentre molecular combination? Contrast allene direction constraints with nitrate delocalisation.

H4

Assembly checkpoint

Guided

Reserve the last judgement for yourself. Try this boundary/transfer task: Explain allene perpendicularity without

using the phrase orbital already occupied.

Answer and working

Hide

Reasoning route

1 Understand

Construct the ethene baseline.

Uses

Ethene is planar; propadiene has terminal CH₂ groups in perpendicular planes. Build a connected account from each carbon's local sigma framework, the required pi overlaps and the p-orbital directions. Explain why copying the planar ethene sketch to allene fails.

Why valid: Each carbon uses three planar sigma directions and one residual perpendicular p.

Result: One pair of parallel p functions; a coplanar sigma framework.

2 Connect

Key move

Reallocate the central carbon for allene.

Uses One pair of parallel p functions; a coplanar sigma framework.

Why valid: Its two sigma-neighbour directions use sp, leaving two perpendicular p functions.

Result: Two independent central p directions rather than ethene's single p direction.

3 Calculate

Match each terminal p to its corresponding central direction.

Uses Two independent central p directions rather than ethene's single p direction.

Why valid: The terminal sigma plane is perpendicular to its p, so distinct orthogonal p alignments constrain distinct planes.

Result: Perpendicular terminal CH₂ planes.

4 Interpret

Reject both a copied planar sketch and the Pauli explanation.

Uses Perpendicular terminal CH₂ planes.

Why valid: The supplied localized construction requires orthogonal directions; Pauli limits state occupancy, not all multicentre participation of one basis function.

Result: The plane relation follows orbital alignment in this model, not a universal one-bond-per-orbital prohibition.

Allene central sp accounting retains two perpendicular p functions. Terminal sigma planes are perpendicular to their respective p functions. Do not claim the first p orbital is filled by one bond and forbidden from multicentre participation by Pauli.

Independent check

Changed check: a delocalised nitrate p function participates in a multicentre description, falsifying the claimed one-AO-one-bond prohibition.

Test the rule on a changed case

Does forming a pi contribution make an atomic p function unavailable to every multicentre molecular combination? Contrast allene direction constraints with nitrate delocalisation.

Your prediction and reason

Q5 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Local hybridisation models, rotational overlap symmetry, and conjugated pathways

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Hybridisation models, symmetry, and conjugated pathways

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For the benchmark, use these supplied simplified observations, not an exam-source claim: CO₂ is linear with two equivalent C–O bonds; NO₃[–] is planar with three equivalent N–O bonds; NH₃ is pyramidal with three equivalent N–H bonds. Assess the claim 'Equivalent bonds imply sp² at the central atom.' Use counterexamples and valid warrants, and distinguish equivalence, electron-domain count and hybridisation as a model description.

Common wrong route · open if you want a warning

–

Common wrong route

Bond equivalence by itself implies sp².

Need a hint? · guided support

–

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Evaluate bond equivalence as a criterion rather than as a conclusion.

H1 Representation

Guided

Equivalent bond properties and local orbital-basis selection are separate statements.

H2 First move

Guided

Test the rule on equivalent-bond examples with different central domain counts.

H3 Equation / formal model

Guided

Test your proposed route on this changed case before finalising: Apply the proposed equality-implies-sp2 rule to CH4 and CO2, then compare it with a domain inventory.

H4 Assembly checkpoint

Guided

Reserve the last judgement for yourself. Try this boundary/transfer task: Write a qualified hybridisation procedure whose inputs exclude bond equivalence as a sufficient test.

Need the concept again?

Local hybridisation models,
rotational overlap symmetry,
and conjugated pathways

Answer and working

Hide

Reasoning route

1 Understand

List the supplied local inventories beside equivalence.

Uses

For the benchmark, use these supplied simplified observations, not an exam-source claim: CO₂ is linear with two equivalent C–O bonds; NO₃[–] is planar with three equivalent N–O bonds; NH₃ is pyramidal with three equivalent N–H bonds. Assess the claim 'Equivalent bonds imply sp² at the central atom.' Use counterexamples and valid warrants, and distinguish equivalence, electron-domain count and hybridisation as a model description.

Why valid: CO₂ C has two domains; nitrate N three; NH₃ N four although each case has equivalent central bonds.

Result: Two, three and four local totals with equivalence held true.

2 Connect

Key move

Use CO₂ or NH₃ as a counterexample.

Uses Two, three and four local totals with equivalence held true.

Why valid: One case satisfying the premise but failing the claimed sp² consequence refutes a universal implication.

Result: CO₂ uses introductory sp; NH₃ sp³-like; the claim is false.

3 Calculate

Separate model labels from observable bond equality.

Uses CO₂ uses introductory sp; NH₃ sp³-like; the claim is false.

Why valid: Domain count and orbital allocation are different layers from measured equivalence.

Result: Only nitrate's three-direction planar framework supports the local sp² description here.

4 Interpret

Change to CH₄ as an independent test.

Uses

Only nitrate's three-direction planar framework supports the local sp² description here.

Why valid: Equivalent C–H bonds coexist with four local directions and an sp³ model.

Result: The rejected equivalence-only rule fails a fresh fourth case.

Equivalent bonds do not imply sp^2 : CO_2 carbon is sp , nitrate N sp^2 , and NH_3 N sp^3 -like in their introductory models.

Independent check

Changed check: CH_4 has equivalent C-H bonds but four carbon directions, contradicting equality-implies- sp^2 .

Test the rule on a changed case

Apply the proposed equality-implies- sp^2 rule to CH_4 and CO_2 , then compare it with a domain inventory.

Your prediction and reason

Q6 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Local hybridisation models, rotational overlap symmetry, and conjugated pathways

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Hybridisation models, symmetry, and conjugated pathways

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For propyne, $CH_3-C\equiv CH$, assign the introductory hybridisation of each carbon and count the sigma and pi

Common wrong route · open if you want a warning

—

bonds in the entire molecule.
Show a check that
distinguishes the triple bond's
components from the local
domain count.

Common wrong route

Each triple-bond line is a sigma bond.

Need a hint? · guided support

—

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Assign each carbon locally and count sigma/pi components across the whole structure.

H1 Representation

Guided

A triple bond supplies one sigma plus two pi components.

H2 First move

Guided

Number the three carbons; inventory local directions and count every connected atom pair once for sigma.

Need the concept again?

Local hybridisation models,
rotational overlap symmetry,
and conjugated pathways

Answer and working

Hide

Reasoning route

1 Understand

Classify each carbon from its own neighbours in propyne.

Uses

For propyne, $\text{CH}_3\text{-C}\equiv\text{CH}$, assign the introductory hybridisation of each carbon and count the sigma and pi bonds in the entire molecule. Show a check that distinguishes the triple bond's components from the local domain count.

Why valid: CH_3 has four directions; both triple-bond carbons have two sigma-neighbour directions.

Result: C labels in order: sp^3 , sp , sp .

2 Connect

Key move

Count sigma components by atom connectivity.

Uses C labels in order: sp^3 , sp , sp .

Why valid: Each connected atom pair has one sigma component regardless of bond order.

Result: Four C-H plus two C-C sigma components = six.

3 Calculate

Count the extra components of the triple bond.

Uses Four C-H plus two C-C sigma components = six.

Why valid: A triple bond contains one sigma and two pi components in the introductory account.

Result: Two pi bonds in the whole molecule.

4 Interpret

Reconcile local and whole-molecule inventories.

Uses Two pi bonds in the whole molecule.

Why valid: The triple bond connects one neighbour at either end even though it has three components.

Result: Six sigma/two pi globally; two local directions at each sp carbon, with no contradiction.

Propyne $\text{CH}_3\text{-C}\equiv\text{CH}$ has carbon labels sp^3 , sp , sp ; four C-H plus two C-C sigma bonds gives six sigma, and the triple bond gives two pi.

Independent check

Independent electron ledger: C_3H_4 has 16 valence electrons; six sigma pairs plus two pi pairs account for sixteen.

Test the rule on a changed case

Count sigma bonds in propyne, then in ethyne. Do double/triple bond lines each count as additional sigma bonds?

Your prediction and reason

Q7 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Delocalised pi systems and stationary resonance hybrids

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Delocalised pi systems in planar oxyanions

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

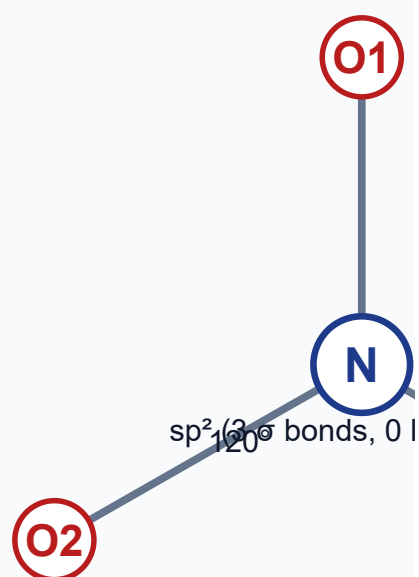
ATTEMPT FIRST

The benchmark supplies that carbonate, CO_3^{2-} , is planar and has three equivalent C–O bonds; its three conventional Lewis contributors place one C=O bond in different positions. Construct one

Representation

qualitative description of its sigma framework and delocalised pi system. State why no single contributor should be read as the complete physical picture.

Stage 1: Trigonal Planar Sigma Framework



Planar sigma frame

Common wrong route • open if you want a warning

Common wrong route

A delocalised carbonate pi space contains only one pair; formal charge averages are measured partial charges.

Need a hint? • guided support

HINT LADDER • REVEAL ONLY WHAT YOU NEED

Need the concept again?

Delocalised pi systems and stationary resonance hybrids

Guided support

H0 Key concept

Guided

Include the terminal-oxygen contributions when describing carbonate pi occupancy.

H1 Representation

Guided

Formal charge averages and physical partial charges have different meanings.

H2 First move

Guided

Keep sigma/in-plane and perpendicular-p ledgers separate; identify what a contributor average actually averages.

H3

Equation / formal model

Guided

Test your proposed route on this changed case before finalising: How many carbonate pi-space electrons arise from one C=O pi pair plus two perpendicular O lone pairs? Does averaging

formal charges measure
physical partial charges?

H4

Assembly checkpoint

Guided

Reserve the last
judgement for yourself.

Try this

boundary/transfer task:

Explain why equal bond
lengths do not uniquely
determine atomic partial
charges.

Answer and working

Hide

Reasoning route

1 Understand

Preserve carbonate connectivity across contributors.

Uses

The benchmark supplies that carbonate, CO_3^{2-} , is planar and has three equivalent C–O bonds; its three conventional Lewis contributors place one C=O bond in different positions. Construct one qualitative description of its sigma framework and delocalised pi system. State why no single contributor should be read as the complete physical picture.

Why valid: All contributors use one C centre and three O neighbours with total charge -2.

Result: Three planar C–O sigma directions.

2 Connect

Key move

Construct the delocalised pi basis and electron ledger.

Uses Three planar C-O sigma directions.

Why valid: Central C plus three O p functions span the pi space; a contributor has one pi pair plus p-oriented pairs at two singly bonded O sites.

Result: Four p functions and six pi-space electrons.

3 Calculate

Interpret contributor charge bookkeeping.

Uses Four p functions and six pi-space electrons.

Why valid: Three equivalent contributors average the two formal negative O charges across three sites.

Result: Average formal charge $-2/3$ per O; this is not a uniquely measured partial charge.

4 Interpret

Check why one fixed contributor is incomplete.

Uses

Average formal charge $-2/3$ per O; this is not a uniquely measured partial charge.

Why valid: A permanently selected C=O position conflicts with the supplied three equivalent C-O bonds.

Result: Use one delocalised description, preserving nuclei, total charge and electron count.

Carbonate has three in-plane sigma bonds and four perpendicular p functions holding six pi-space electrons. Average formal O charge $-2/3$ is contributor bookkeeping, not a uniquely measured atomic partial charge.

Independent check

Independent ledger: CO_3^{2-} has 24 valence electrons; three sigma pairs and six in-plane O lone pairs use eighteen, leaving six in the pi space.

Test the rule on a changed case

How many carbonate pi-space electrons arise from one C=O pi pair plus two perpendicular O lone pairs? Does averaging formal charges measure physical partial charges?

Your prediction and reason

Q8 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Local hybridisation models, rotational overlap symmetry, and conjugated pathways

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Hybridisation models, symmetry, and conjugated pathways

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Compare rotation about the central C–C bond in ethane with twisting about the C=C bond in ethene. Use the sigma and pi overlap arrangements to explain which overlap is altered by each motion. Do not infer a numerical energy barrier from a sketch.

Common wrong route · open if you want a warning

–

Common wrong route

A fixed central sigma axis means every molecular interaction is unchanged.

Need a hint? · guided support

–

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Separate persistence of the axial sigma connection from claims about all electronic interactions.

H1 Representation Guided

Relative p alignment affects pi overlap; one axial sketch is not a total-energy decomposition.

H2 First move Guided

Rotate relative groups while holding the central bond axis fixed; state only what this model supports.

H3 Equation / formal model

Guided

Test your proposed route on this changed case before finalising: During ethane rotation, does an unchanged C-C axis guarantee unchanged orientation of every neighbouring orbital? Compare ethene p alignment.

H4 Assembly checkpoint Guided

Reserve the last judgement for yourself. Try this boundary/transfer task: Explain the sigma/pi contrast without assigning a universal torsional energy cause.

Need the concept again?

Local hybridisation models,
rotational overlap symmetry,
and conjugated pathways

Answer and working

Hide

Reasoning route

1

Understand

Specify the axial C-C sigma reference in both molecules.

Uses

Compare rotation about the central C-C bond in ethane with twisting about the C=C bond in ethene. Use the sigma and pi overlap arrangements to explain which overlap is altered by each motion. Do not infer a numerical energy barrier from a sketch.

Why valid: Rotation about that axis can preserve the axial sigma overlap.

Result: A sigma connection can persist during torsion.

2

Connect

Key move

Identify the extra ethene pi alignment requirement.

Uses A sigma connection can persist during torsion.

Why valid: Its parallel residual p functions overlap laterally; relative group twist changes that alignment.

Result: Ethene twist alters pi overlap; ethane has no corresponding C-C pi component.

3

Calculate

Separate rigid rotation from relative torsion.

Uses Ethene twist alters pi overlap; ethane has no corresponding C-C pi component.

Why valid: Only relative orientation about the bond changes the participating p relation.

Result: Rotating the whole molecule does not reproduce twisting one group.

4 Interpret

Bound the energy inference.

Uses Rotating the whole molecule does not reproduce twisting one group.

Why valid: Other interactions, including hyperconjugation, can change even when the central sigma connection remains.

Result: No numerical barrier or exclusively steric total-energy law follows from the sketch.

Ethane axial C-C sigma bonding can persist through torsion; ethene pi alignment changes under twist. Other interactions, including hyperconjugation, can change; an exclusively steric total-energy account is unjustified.

Independent check

Changed check: a fixed central sigma direction does not hold every C-H/donor-acceptor orientation fixed.

Test the rule on a changed case

During ethane rotation, does an unchanged C-C axis guarantee unchanged orientation of every neighbouring orbital? Compare ethene p alignment.

Your prediction and reason

Q9 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding

Concept: Local hybridisation models, rotational overlap symmetry, and conjugated pathways

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

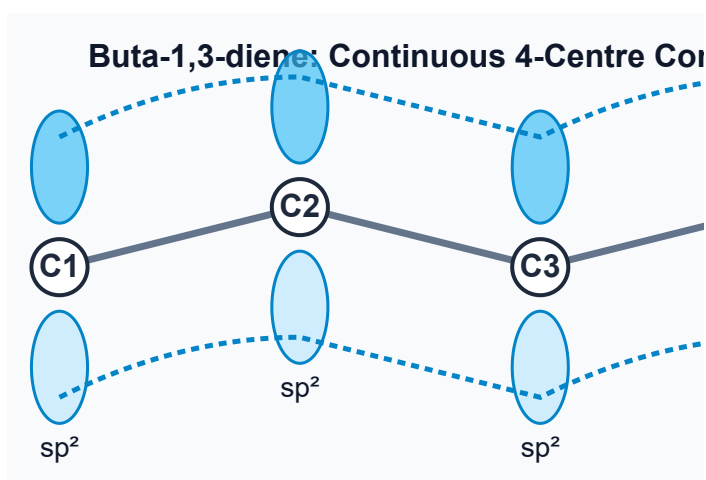
Family: Hybridisation models, symmetry, and conjugated pathways

Question type: Constructed response

Source: Owner-supplied question

ATTEMPT FIRST

For $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ in an approximately planar conformation, identify the continuous sequence of p orbitals and contrast it with $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}=\text{CH}_2$. Explain which structural feature interrupts that pathway in the second molecule.

Representation

Common wrong route • open if you want a warning

Common wrong route

Spatial closeness repairs an interrupted p pathway.

Need a hint? • guided support

HINT LADDER • REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Need the concept again?

Local hybridisation models, rotational overlap symmetry, and conjugated pathways

Trace the continuous aligned p pathway rather than counting all double bonds.

H1 Representation

Guided

An intervening saturated sp^3 centre breaks the ordinary contiguous pi pathway in this model.

H2 First move

Guided

Mark p-bearing centres and inspect adjacency before declaring conjugation.

H3

Equation / formal model

Guided

Test your proposed route on this changed case before finalising:
Compare $CH_2=CH-CH=CH_2$ with $CH_2=CH-CH_2-CH=CH_2$. Does spatial closeness alone repair the missing p pathway?

H4

Assembly checkpoint

Guided

Reserve the last judgement for yourself.
Try this boundary/transfer task:
Specify a structural change that restores the pathway and a model limitation.

Answer and working

Hide

Reasoning route

1 Understand

Trace consecutive p-bearing carbon sites in butadiene.

Uses

For $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ in an approximately planar conformation, identify the continuous sequence of p orbitals and contrast it with $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}=\text{CH}_2$.
Explain which structural feature interrupts that pathway in the second molecule.

Why valid: An approximately planar conjugated chain supplies a p function at each of its four sp^2 sites.

Result: A four-site contiguous aligned p pathway.

2 Connect

Key move

Trace the second molecule using the same local test.

Uses A four-site contiguous aligned p pathway.

Why valid: Its intervening saturated CH_2 uses localized sp^3 accounting, with no ordinary residual p in this model.

Result: The central spacer interrupts the contiguous pathway.

3 Calculate

Distinguish adjacency from spatial proximity.

Uses The central spacer interrupts the contiguous pathway.

Why valid: Moving separated fragments near each other does not remove the saturated node from their bonding path.

Result: The ordinary through-chain conjugation criterion remains unmet.

4 Interpret

State the model boundary explicitly.

Uses The ordinary through-chain conjugation criterion remains unmet.

Why valid: Absence of the chosen contiguous p path does not exclude every possible quantum interaction.

Result: No absolute zero-interaction claim; reassess if the spacer becomes p-bearing and suitably aligned.

Butadiene has four contiguous p-bearing centres; a central saturated CH₂ in separated pentadiene interrupts this introductory pathway. This is not a proof of zero interaction of every quantum-mechanical kind.

Independent check

Changed check: replacing the saturated spacer with an appropriate unsaturated p-bearing centre restores a contiguous path.

Test the rule on a changed case

Compare CH₂=CH-CH=CH₂ with CH₂=CH-CH₂-CH=CH₂. Does spatial closeness alone repair the missing p pathway?

Your prediction and reason

Subject: Chemistry

Topic: Chemical Bonding

Concept: Local hybridisation models, rotational overlap symmetry, and conjugated pathways

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Hybridisation models, symmetry, and conjugated pathways

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A student assigns one hybridisation label to an entire molecule. Use propyne and allene to test this procedure and devise a more reliable local procedure. State the electron-domain information and orbital-consistency checks your procedure needs.

Common wrong route · open if you want a warning

—

Common wrong route

A whole molecule has one hybridisation label.

Need a hint? · guided support

—

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Synthesize a local procedure that also checks inter-centre compatibility.

H1 Representation

Guided

Local count, residual p directions and model applicability have different jobs.

H2 First move

Guided

Apply the same ordered inventory to each atom, then check neighbouring

overlaps; preserve boundaries.

H3 Equation / formal model

Guided

Test your proposed route on this changed case before finalising: Can a whole molecule have one hybridisation label? Run a local ledger for allene and propyne.

H4 Assembly checkpoint

Guided

Reserve the last judgement for yourself. Try this boundary/transfer task: State the evidence that would require revisiting a localized lone-pair assignment.

Need the concept again?

Local hybridisation models,
rotational overlap symmetry,
and conjugated pathways

Answer and working

Hide

Reasoning route

1 Understand

Test one-label-per-molecule on propyne.

Uses

A student assigns one hybridisation label to an entire molecule. Use propyne and allene to test this procedure and devise a more reliable local procedure. State the electron-domain information and orbital-consistency checks your procedure needs.

Why valid: Its carbon local inventories differ: four, two, two directions.

Result: sp^3 , sp , sp ; a single whole-molecule label fails.

2 Connect

Key move

Test the procedure on allene.

Uses sp^3 , sp , sp ; a single whole-molecule label fails.

Why valid: The terminal and central inventories are three, two, three directions.

Result: sp^2 , sp , sp^2 ; two perpendicular central p directions must remain.

3 Calculate

Build the replacement atom-local procedure.

Uses sp^2 , sp , sp^2 ; two perpendicular central p directions must remain.

Why valid: Count neighbours and lone pairs, conserve each local valence basis, then assign residual p roles.

Result: A label and orbital ledger for each atom.

4 Interpret

Check neighbouring orientation and applicability.

Uses A label and orbital ledger for each atom.

Why valid: Adjacent p functions must support the intended pi account; delocalisation or unusual occupancy can require a richer model.

Result: A qualified local framework with explicit compatibility and stop-for-more-evidence checks.

Count local sigma directions and lone pairs, conserve the valence basis, check residual p orientation and neighbouring compatibility, then qualify applicability. Propyne/allene classifications are atom-local.

Independent check

Changed check: the procedure must return allene $sp^2/sp/sp^2$ and propyne $sp^3/sp/sp$ while conserving four carbon basis functions at each site.

Test the rule on a changed case

Can a whole molecule have one hybridisation label? Run a local ledger for allene and propyne.

Your prediction and reason